



JAL1501 MACHINE LEARNING

UNIT III ENSEMBLE TECHNIQUES AND UNSUPERVISED LEARNING

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SYLLABUS

Combining multiple learners: Model combination schemes, Voting, Ensemble Learning bagging, boosting, stacking.

Unsupervised learning: K-means, Instance Based Learning: KNN, Gaussian mixture models and Expectation maximization.



ENSEMBLE LEARNING

“COMBINING MULTIPLE LEARNERS”

A powerful machine learning technique that involves combining the predictions from multiple models to improve overall performance.

Ensemble Learning

- Ensemble learning in machine learning combines multiple individual models (often called "weak learners") to create a more powerful and accurate predictive model, known as a "strong learner."
- Ensemble learning is a method where we use many small models instead of just one. Each of these models may not be very strong on its own.
- The core idea is that the collective intelligence of multiple models can mitigate individual model weaknesses, reduce bias and variance, and improve overall generalization performance.
- Key ensemble learning techniques include the following:
 - Voting
 - Bagging (Bootstrap Aggregating)
 - Boosting
 - Stacking

Combining Multiple Learners

- A powerful machine learning technique that involves combining the predictions from multiple models to improve overall performance
- When designing a ML model, we generally make few choices – like parameters, training data and representation → this causes some variance in the performance.
- Each learning algorithm dictates a certain model that comes with a set of assumptions. This inductive bias leads to error if the assumptions do not hold for the data.
- Different learning algorithms have different accuracies.
- The no free-lunch theorem in ML asserts that no single learning algorithm always achieves the best performance in any domain → emphasizes that every algorithm has strengths and weaknesses, and its effectiveness is highly dependent on the specific problem and data it is applied to.
- They can be combined to attain higher accuracy.

Combining Multiple Learners

- Data fusion → the process of fusing multiple records representing the same real-world object into a single, consistent, and clean representation.
- Fusion of data for improving prediction accuracy and reliability is an important problem in machine learning.
- Combining different models is done to improve the performance of deep learning models.
- Building a new model by **combination** requires less time, data, and computational resources.
- The most common method to combine models is by averaging multiple models, where taking a weighted average improves the accuracy.
- Base-learners → the individual models that make up the ensemble. Base learners can be of the same type (homogeneous ensemble) or different types (heterogeneous ensemble). For example, an ensemble could consist of multiple decision trees or a combination of decision trees, support vector machines, and neural networks.

Combining Multiple Learners

Generating Diverse Learners:

- ***Different Algorithms:*** Different learning algorithms can be used to train different base-learners. Different algorithms make different assumptions about the data and lead to different classifiers.
- ***Different Hyper-parameters:*** The same learning algorithm can be used but with different hyper – parameters.
- ***Different Input Representations:*** Different representations make different characteristics explicit allowing better identification.
- ***Different training sets:*** Training different base-learners by different subsets of the training set.

MODEL COMBINATION SCHEMES

Model Combination Schemes

- Involve integrating multiple models to improve overall performance, accuracy, and robustness in machine learning tasks.
- Different methods used for generating final output for multiple base learners:
 - Multiexpert combination method
 - Multistage combination method
- Common methods: bagging, boosting, stacking, ensemble averaging
- Aim to leverage the strengths of different models while minimizing their weaknesses.

Multiexpert Combination Method

- A technique in ML and decision-making where multiple experts (or models) are combined to make a more accurate prediction or decision than any single expert could achieve alone.
- a) Global approach (Learner fusion) : given an input, all base - learners generate an output and all these outputs are used → Eg. Techniques such as voting and stacking
- b) Local approach (Learner selection) : in a mixture of experts, there is a gating model, which looks at the input and chooses one (or every few) of the learners as responsible for generating the output.

Model Combination Schemes

Multistage Combination Method

- An advanced ensemble learning technique where multiple learners (models) are combined in a sequential manner across different stages.
- This method integrates various models in a layered approach, where the output of one stage serves as the input for the next.
- The goal → to **refine and improve predictions progressively**, leveraging the strengths of different models at each stage.
- Use a serial approach where the next multistage combination base – learner is trained with or tested on only the instances where the previous base – learners are not accurate enough.
- Eg. Techniques such as boosting and stacking.

ENSEMBLE LEARNING

Ensemble Learning

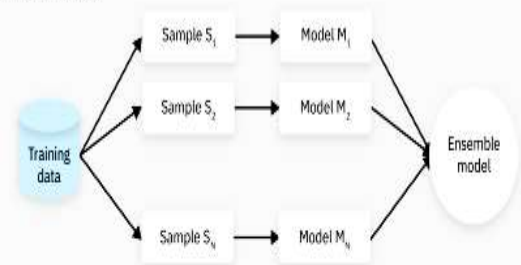
- The simplest way to combine multiple classifiers
- It is a ML technique that aggregates two or more learners (e.g. regression models, neural networks) in order to produce better predictions.
- An ensemble model combines several individual models to produce more accurate predictions than a single model alone
- These techniques are widely used in machine learning to build more robust, accurate, and generalizable models.
- Principle → a collectivity of learners yields greater overall accuracy than an individual learner.
- Base learner / base model / base estimator → refers to the individual model or models used to in ensemble algorithms.
- Weak models or learners → those that perform little better than random guessing → eg. For binary classification problems, fifty percent accuracy achieved.
- Strong models or learners → achieve excellent predictive performance → eg. in binary classification greater than or equal to eighty percent accuracy achieved.

Ensemble Learning

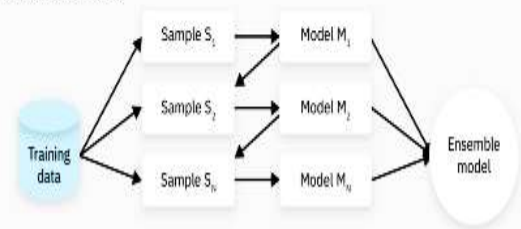
Types of ensemble learning techniques

- Categorised into two groups → parallel and sequential methods
- Parallel methods → train base learners in parallel and independent of one another.
 - Homogenous parallel ensembles use the same base learning algorithm to produce all of the component base learners.
 - Heterogenous parallel ensembles use different algorithms to produce base learners
- Sequential methods → train a new base learner so that it minimizes errors made by the previous model trained in the preceding step → construct base models sequentially in stages.

Parallel ensembles



Sequential ensembles



Ensemble Learning

Ensemble learning techniques

- | | | | |
|---|---|--|--|
| <ul style="list-style-type: none"> ▪ Bagging ▪ Boosting ▪ Stacking ▪ Blending | } | systematic ensemble frameworks with well-defined algorithms | |
| <ul style="list-style-type: none"> ▪ Voting ▪ Weighted average | } | | ensemble strategies for combining model predictions |

VOTING

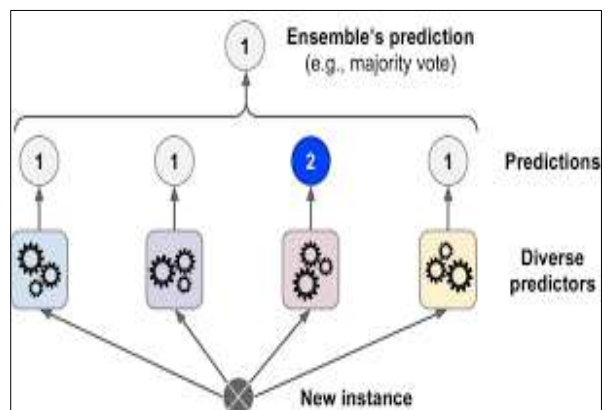
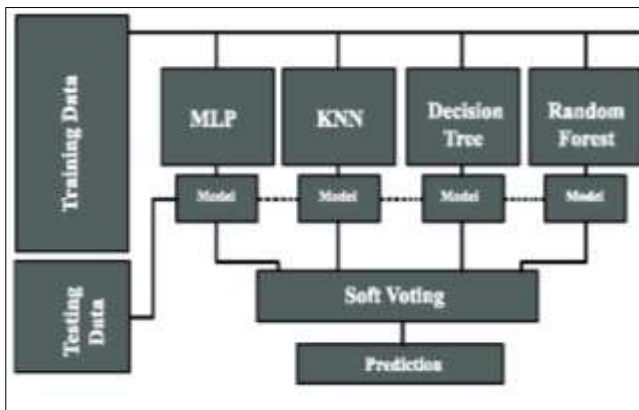
Voting

- Voting is an ensemble method.
- The simplest way to combine multiple classifiers - combines the predictions of multiple models (often different types) to make a final prediction.
- Corresponds to taking a linear combination of the learners.
- Combines the predictions of different models by majority voting or averaging.

Voting

- For regression, a voting ensemble → makes a prediction that is the average of multiple other regression models.
- In classification,
 - A **hard voting (majority voting)** ensemble → sums the votes for crisp class labels from other models and predicts the class with the **most votes**
 - If models 1, 2, 3 assign 'YES' and model 4 assigns 'NO', 'YES' is chosen
 - Weighted majority voting → an extension of this technique that gives greater weight to certain learner's predictions over others
 - A **soft voting (weighted voting)** ensemble → **sums** the predicted probabilities for class labels and predicts the class label with the largest sum probability estimate.
 - If model 1 gives $P(\text{YES})=0.64$, $P(\text{NO})=0.12$ and model 2 gives $P(\text{YES})=0.52$, $P(\text{NO})=0.37$, YES is chosen (because $0.64+0.52 > 0.12+0.37$)

Voting



Voting

- In this method, the first step is to create multiple classification/regression models using some training dataset.
- Each base model can be created using different splits of the same training dataset and same algorithm, or using the same dataset with different algorithms, or any other methods.
- When combining multiple independent and diverse decisions each of which is at least more accurate than random guessing, random errors cancel each other out, and correct decisions are reinforced. Human ensembles are demonstrably better.
- Use a single, arbitrary learning algorithm but manipulate training data to make it learn multiple models.

Voting

How Voting Works

- **Model Selection:** First, you choose a set of models to include in the ensemble. These models can be different types (e.g., decision trees, logistic regression, SVMs) or the same type but with different hyperparameters.
- **Training:** Each model in the ensemble is trained independently on the same training data.
- **Prediction:** When a new input is given, each model makes its prediction.
 - In majority voting, the final prediction is based on the majority class predicted by the models.
 - In weighted voting, the final prediction is based on the weighted average of the probabilities predicted by each model.

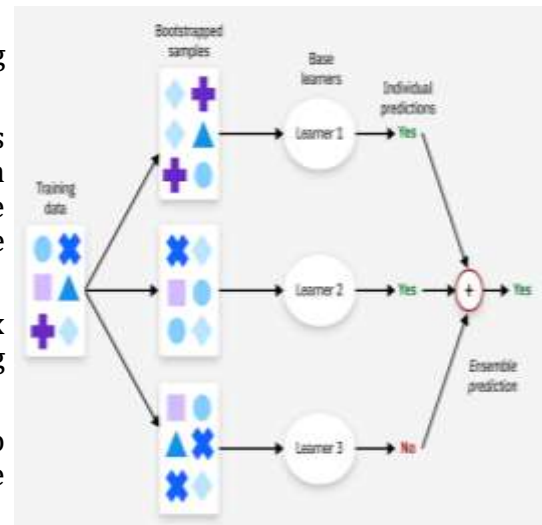
BAGGING

Bagging

- A homogenous parallel method, also called as **Bootstrap Aggregating**
- Involves training multiple versions of a model on **different subsets** of the training data.
- Uses a technique called **bootstrap resampling** to derive multiple new datasets from one initial training dataset in order to train multiple base learners.
- These subsets are created by randomly sampling with replacement from the original dataset (bootstrapping).
- The final prediction is made by **averaging the predictions of all models** (for regression) or by taking the **majority vote** (for classification).
- Reduces variance by training models on different subsets of the data and averaging their predictions and prevents overfitting.
- Example: Random Forest is a classic example of Bagging → where multiple decision trees are trained on different bootstrapped samples of the data.

Bagging

- **How does this work?**
- Assume a training dataset contains n training examples.
- Bootstrap resampling copies n data instances from that set into a new subsample dataset, with some initial instances appearing more than once and others excluded entirely → These are **bootstrap samples (or) bootstrap replicates**
- Repeating this process x times produces x iterations of the original dataset, each containing n samples from the initial set.
- Each iteration of the initial set is then used to train a separate base learner with the same learning algorithm.

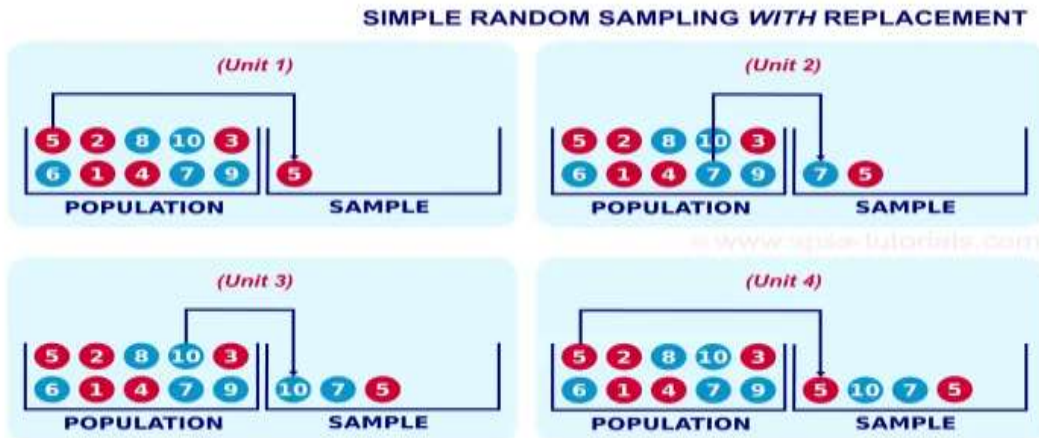


Bagging

- Each model trained on the **bootstrap replicate** of the original training set.
- [Suppose the training set consists of n examples. Then a bootstrap replicate is a new training set that also consists of n examples, and which is formed by repeatedly selecting uniformly at random and with replacement n examples from the original training set. This means that the same example may appear multiple times in the bootstrap replicate, or it may appear not at all.
- For given a training set of size n , create m samples of size n by drawing n examples from the original data, with replacement.]
- Each bootstrap sample will on average contain **63.2% of the unique training examples**, the rest are replicates. It combines the m resulting models using simple majority vote.
- **Sampling with replacement** ensures each bootstrap is independent from its peers, as it does not depend on previous chosen samples when sampling
- Decreases error by decreasing the variance in the results due to unstable learners, algorithms (like decision trees) whose out put can change dramatically when the training date is slightly changed

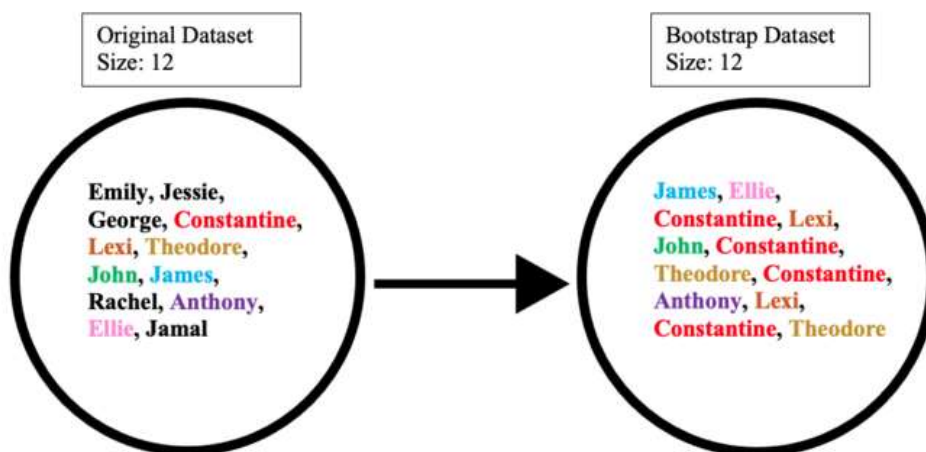
Bagging

- Sampling with replacement → has replicated rows



Bagging

- A sample **bootstrap replicate** of the original training set.



Bagging

Pseudocode:

1. Given Training data $(x_1, y_1) \dots\dots (x_m, y_m)$
2. For $t = 1, \dots T$:
 - a. Form bootstrap replicate dataset S_t by selecting m random examples from the training set with replacement.
 - b. Let h_t be the result of training base learning algorithm on S_t .
3. Output combined classifier : $H(x) = \text{majority}(h_1(x) \dots\dots h_T(x))$

Bagging

Advantages of Bagging :

- Reduces over – fitting of the model.
- Handles higher dimensionality data very well.
- Maintains accuracy for missing data.
- Many weak learners aggregated typically outperform a single learner over the entire set, and have less overfit.
- Reduces variance in high-variance low-bias weak learner (Eg. DT, NN, SVM, KNN, RF), which can improve efficiency
- Can be performed in parallel, as each separate bootstrap can be processed on its own before aggregation.

Disadvantages of Bagging :

- Since final prediction is based on the mean predictions from subset trees, it won't give precise values for the classification and regression model
- For a weak learner with high bias (Lin Reg, Log Reg, NB, KNN, SVM, DT), bagging will also carry high bias into its aggregate
- Loss of interpretability of a model.
- Can be computationally expensive depending on the data set.

BOOSTING

Boosting

- Boosting is a sequential ensemble technique where each new model is trained to correct the errors made by the previous models.
- Models are added one at a time
- Each new model focuses more on the data points that were misclassified by the earlier models.
- The final prediction is a weighted sum of the predictions from all models.
- Effective in reducing bias and is often used in scenarios where high accuracy is required.
- Reduces bias by sequentially adding models that correct the errors of the previous ones.
- Example: AdaBoost, Gradient Boosting, and XGBoost → popular Boosting algorithms.

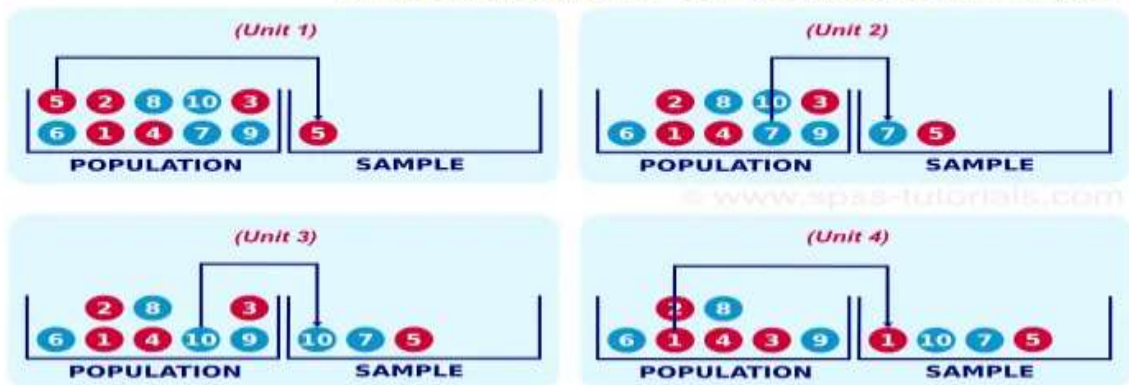
Boosting

- Boosting has many variations - but they all follow the same general procedure.
- Boosting trains a learner on some initial dataset, d .
- The resultant learner is typically weak, misclassifying many samples in the dataset.
- Boosting then samples instances from the initial dataset to create a new dataset (d_2) → prioritizes misclassified data instances from the first model or learner.
- A new learner is trained on this new dataset d_2 .
- Then a third dataset (d_3) is then compiled from d_1 and d_2 , prioritizes the second learner's misclassified samples and instances in which d_1 and d_2 disagree.
- The process repeats n times to produce n learners. Boosting then combines and weights all the learners together to produce final predictions.

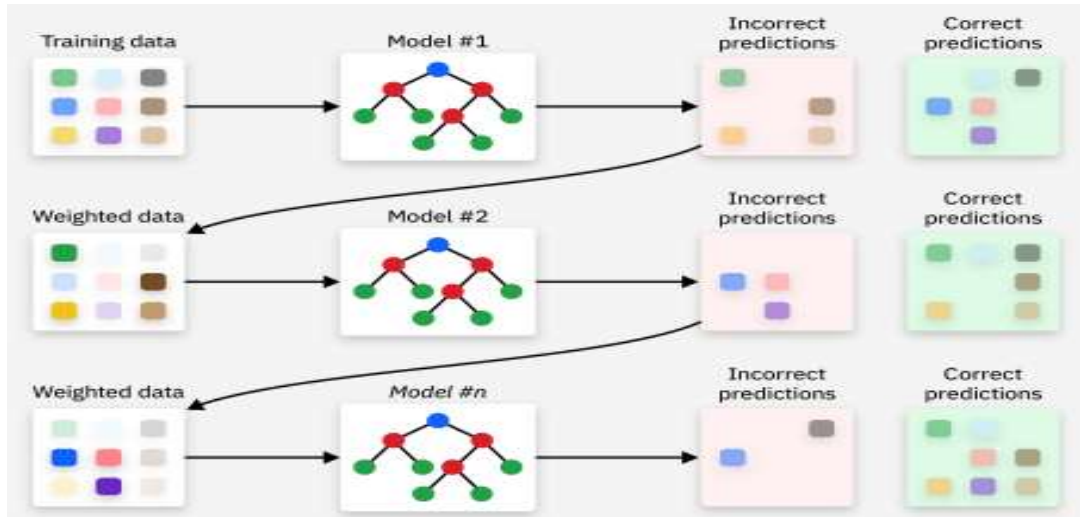
Boosting

- Uses sampling without replacement → has unique rows

SIMPLE RANDOM SAMPLING WITHOUT REPLACEMENT



Boosting



Boosting

Pseudocode for Boosting algorithm:

1. Draw a random subset of training samples d_1 without replacement from the training set D to train a weak learner C_1
2. Draw second random training subset d_2 without replacement from the training set add 50 percent of the samples that were previously falsely classified / misclassified to train a weak learner C_2
3. Find the training samples d_3 in the training set D on which C_1 and C_2 disagree to train a third weak learner C_3
4. Combine all the weak learners via majority voting.

Boosting

- Boosting has many variations - differ in how they prioritize erroneously predicted data instances when creating a new dataset.
- **Adaptive boosting (AdaBoost)** weights model errors → when creating a new iteration of a dataset for training the next learner, AdaBoost adds weights to the previous learner's misclassified samples, causing the next learner to prioritize those misclassified samples.
- **Gradient boosting** uses residual errors (actual value – estimated value) when training new learners → uses residual errors from a previous model to set target predictions for the next model → thus attempts to close the gap of error left by one model.
- **XGBoost** - Extreme Gradient Boosting → combines the predictions of multiple weak models to produce a stronger prediction

Boosting

Adaptive boosting (AdaBoost)

- A machine learning meta – algorithm used in conjunction with any other types of learning algorithms to improve their performance.
- It can be used **to learn weak classifiers** and final classification based on weighted vote of weak classifiers.
- Has good generalization properties.
- Initially, all weights are set equally, but after each round the weights of incorrectly classified examples are increased so that those observations that the previously classifier poorly predicts receive greater weight on the next iteration.

Boosting

Algorithm for Adaboost:

1. For a dataset with N training data points instances, initialize the weights as $1 / N$.
2. For K iterations
 - a. Train weak classifier M_k .
 - b. Calculate the error rate for every weak classifier M_k on the training dataset
 - c. Update data point weight for each data point W_i
 - d. Normalize the instance weight so that they can be summed up to 1 using the formula $W_i = W_i / \text{sum}(W)$.
3. The final model $M(X)$ will be an ensemble model which is obtained by combining these weak models weighted by their model weights.

Boosting

Advantages of Adaboost :

- Very simple to implement
- Fairly good generalization
- The prior error need not be known ahead of time.

Disadvantages of Adaboost :

- Suboptimal solution
- Can overfit in presence of noise.

Advantages of Boosting :

- Supports different loss functions.
- Works well with interactions..

Disadvantages of Boosting :

- Prone to over – fitting.
- Requires careful tuning of different hyper – parameters..

Bagging vs Boosting

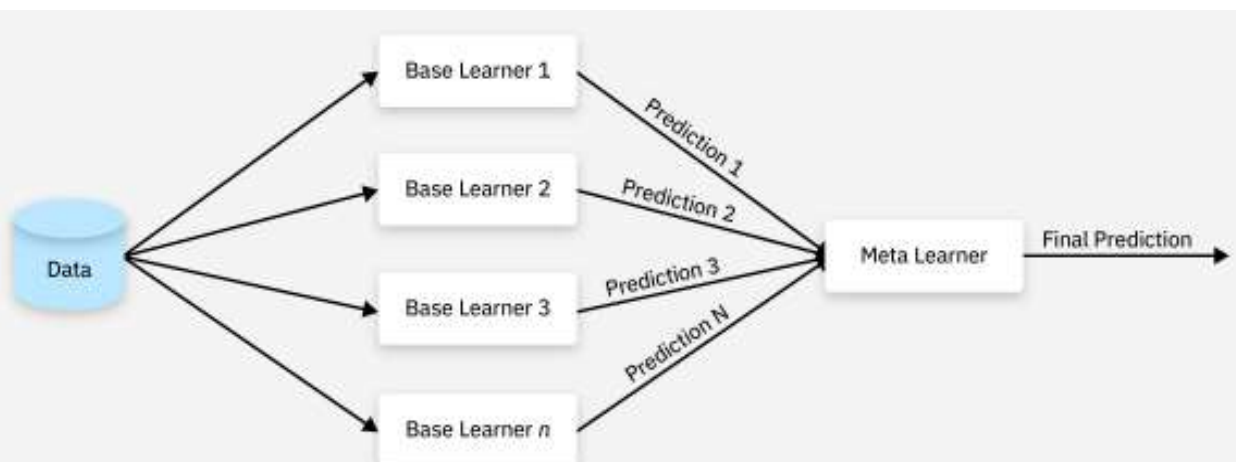
Sl. No.	BAGGING	BOOSTING
1.	Technique that builds multiple homogeneous models from different subsamples of the same training dataset to obtain more accurate predictions than its individual models	A group of algorithms that utilize weighted averages to make weak learning algorithms stronger learning algorithms.
2.	Learns them independently from each other in parallel	Learns them sequentially in very adaptative way
3.	It helps in reducing variance.	It helps in reducing bias and variance.
4.	Every model receives an equal weight.	Models are weighted by their performance.

STACKING

Stacking

- Also called as Stacked Generalization
- A heterogeneous parallel method that demonstrates meta-learning.
- Meta-learning consists of training a meta-learner from the output of multiple base learners
- A more complex ensemble technique
- Multiple models (called base learners) are trained on the same dataset, and their predictions are used as inputs to a meta-model (meta learner), which makes the final prediction.
- The base models can be of different types, and the meta-model learns to combine them in the most effective way.
- Example: Train a decision tree, a logistic regression, and a support vector machine as base learners, and then use a neural network as the meta-learner to combine their predictions.
- Combines the predictions of base models using a meta-model that learns how to best combine them.
- Stacking is powerful when we want to leverage the strengths of different models and improve predictive performance.

Stacking



Stacking

- Using the same dataset to train the base learners and the meta-learner can result in overfitting.
- To avoid this
 - Exclude few data instances from the base learner training data and use the resulting subset of dataset as its test set data (non-overlapping)
 - This test set data becomes training data for the meta-learner.

